

ARNOLD ABRAHAM

SOFTWARE ENGINEER | BACKEND

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Location: Erlangen, Germany (Willing to relocate)

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Experience

Software Engineer – Working Student

Jul 2023 – Jun 2025

Siemens Logistics (Vanderlande), Nuremberg, Germany

- Led the migration of two Java applications from Oracle to PostgreSQL, ensuring data integrity and application stability while reducing database licensing costs.
- Developed backend features and implemented HikariCP connection pooling to improve performance and stability.
- Designed and implemented automated installation workflows for JRE, PostgreSQL, and environment configuration, reducing setup effort by 70%.
- Built a Python-based predictive maintenance pipeline integrating thermal sensor data and ML models, enabling early fault detection and reducing periodic manual inspections by 90%.
- Implemented an asyncio-based Python data acquisition system for concurrent real-time monitoring of industrial drives.
- Collaborated with cross-functional engineering teams to integrate predictive monitoring into production workflows.

Product Engineer

Sept 2021 – Sept 2022

IBS Software, Kochi, India

- Developed Java/Spring-based backend components for an enterprise airline crew scheduling platform used by major global carriers (LATAM Airlines, Qantas), supporting thousands of daily scheduling decisions.
- Designed and implemented complex scheduling and compliance rules, addressing production issues and delivering enhancements based on QA and operational feedback.
- Created and troubleshooted SQL scripts to support backend functionality and ensure data integrity.
- Worked across the full SDLC in an agile environment, including design discussions, unit testing using JUnit and Mockito, peer code reviews, and production support.

Education

MSc Data Science

Oct 2022 – Sept 2025

University of Erlangen–Nuremberg, Germany

- Major in Artificial Intelligence and Machine Learning.
- Thesis: Predictive Maintenance for Electrical Panels: Hotspot Forecasting and Anomaly Detection (Siemens).

B.Tech Computer Science and Engineering

Aug 2017 – Jul 2021

APJ Abdul Kalam Technological University, India

Technical Skills

Languages: Java, Python, SQL

Frameworks: Spring Boot, Spring Framework, Hibernate (JPA)

Cloud & DevOps: AWS, Docker, GitLab CI

Databases: PostgreSQL, Oracle, InfluxDB

Core Competencies: Backend Development, Microservices Architecture, RESTful APIs, Object-Oriented Design, Data Structures & Algorithms, Unit Testing

Languages

English (C1 – Fluent) ▪ German (A1 – Basic)